

CHAPTER 10.1

TANGENT LINES AND ARC LENGTH FOR PARAMETRIC CURVES

Arc Length of Parametric Curves

The following result provides a formula for finding the arc length of a curve from parametric equations for the curve. Its derivation is similar to that of Formula (3) in [Section 6.4](#).

10.1.1 Arc Length Formula for Parametric Curves

If no segment of the curve represented by the parametric equations

$$x = x(t), y = y(t) \text{ for } a \leq t \leq b$$

is traced more than once as t increases from a to b , and if $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are continuous functions for $a \leq t \leq b$, then the arc length L of the curve is given by

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Example 8

page 698 (Anton et al.)

EXERCISE SET 10.1

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at the given point without eliminating the operator:

46. $x = \frac{1}{2}t^2 + 1, y = \frac{1}{3}t^3 - t, t = 2$

We know $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{t^2 - 1}{t}$ and $\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{1}{t} \times \frac{(t)(2t) - (t^2 - 1)}{t^2} = \frac{t^2 + 1}{t^3}$

Hence, $\left. \frac{dy}{dx} \right|_{t=2} = \frac{3}{2}$ and $\left. \frac{d^2y}{dx^2} \right|_{t=2} = \frac{5}{8}$

48. $x = \sinh t, y = \cosh t, t = 0$

We know $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\sinh t}{\cosh t} = \tanh t$ and $\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{\operatorname{sech}^2 t}{\cosh t} = \frac{1}{\cosh^3 t}$

Hence, $\left. \frac{dy}{dx} \right|_{t=0} = 0$ and $\left. \frac{d^2y}{dx^2} \right|_{t=0} = 1$

50. $x = \cos \phi, y = 3 \sin \phi, \phi = 5\pi/6$

We know $\frac{dy}{dx} = \frac{dy/d\phi}{dx/d\phi} = \frac{3 \cos \phi}{-\sin \phi} = -\frac{3}{\tan \phi}$ and $\frac{d^2y}{dx^2} = \frac{dy'/d\phi}{dx/d\phi} = \frac{3 \csc^2 \phi}{-\sin \phi} = -\frac{3}{\sin^3 \phi}$

Hence, $\left. \frac{dy}{dx} \right|_{\phi=5\pi/6} = 3\sqrt{3}$ and $\left. \frac{d^2y}{dx^2} \right|_{\phi=5\pi/6} = -24$

52. (a) Find the equation of the tangent line to the curve

$$x = 2t + 4, y = 8t^2 - 2t + 4$$

at $t = 1$ without eliminating the parameter.

(b) Find the equation of the tangent line in part (a) by eliminating the parameter.

Solution: (a) Given that $x = 2t + 4, y = 8t^2 - 2t + 4$

Calculate $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{16t - 2}{2} = 8t - 1$

Therefore, $\left. \frac{dy}{dx} \right|_{t=1} = 7$

Now when $t = 1$, we have $x = 6$ and $y = 10$.

So the required tangent line in point-slope form is

$$y - 10 = 7(x - 6) \Rightarrow y = 7x - 32$$

(b) When $t = \frac{x-4}{2}$, we have

$$y = 8\left(\frac{x-4}{2}\right)^2 - 2\left(\frac{x-4}{2}\right) + 4 = 2(x^2 - 8x + 16) - x + 4 + 4 = 2x^2 - 17x + 40$$

Also, when $t = 1$, we have $x = 6$ and $\left.\frac{dy}{dx}\right|_{x=6} = [4x - 17]_{x=6} = 7$

So the required tangent line in point-slope form is

$$y - 10 = 7(x - 6) \Rightarrow y = 7x - 32$$

Find all values of t at which the parametric curve has (a) a horizontal tangent line and (b) a vertical tangent line:

53. $x = 2\sin t$, $y = 4\cos t$, $0 \leq t \leq 2\pi$

Solution: (a) Calculate $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-4\sin t}{2\cos t} = -2\tan t$

For horizontal tangent lines $\frac{dy}{dt} = 0 \Rightarrow -4\sin t = 0 \Rightarrow \sin t = 0 \Rightarrow t = 0, \pi, 2\pi$

(b) For vertical tangent lines $\frac{dx}{dt} = 0 \Rightarrow 2\cos t = 0 \Rightarrow \cos t = 0 \Rightarrow t = \frac{\pi}{2}, \frac{3\pi}{2}$

Exercise Set 10.1

Find the exact arc length of the curve over the stated interval:

Problems: 65-70

Problem 66

$x = \sqrt{t} - 2$, $y = 2t^{3/4}$, $1 \leq t \leq 16$

Solution: (a) Let $t = u^2$. Then $x = u - 2$, $y = 2u^{3/2}$, $1 \leq u \leq 4$

Therefore, $\sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2} = \sqrt{1 + 9u}$

Hence, arc length $L = \int_1^4 \sqrt{1 + 9u} \, du = \frac{1}{9} \times \frac{2}{3} [(1 + 9u)^{3/2}]_1^4 = \frac{1}{27} (37\sqrt{37} - 10\sqrt{10})$.